

Mohsen Mostafa

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Professional Summary

AI Lead, Research Engineer, and Systems Architect with 7+ years of experience in Data Science, AI/ML, Computer Vision, and Generative AI across healthcare, geospatial, fintech, and automation domains. Specialized in LLMs, RAG, multimodal AI, AI agents, and scalable ML systems using FastAPI, Docker, PyTorch, and Hugging Face. Proven track record of improving model accuracy, reducing latency, and leading AI initiatives from research to production deployment.

Core Skills

Python, Pandas, Dask, Polars, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Hugging Face, Falcon, DeepSeek, StarCoder, Mistral, Llama, Qwen, RAG, ColBERT, MCP, CLIP, YOLOv8, OpenCV, Mask2Former, LiDAR/DSM, FastAPI, Docker, CI/CD, ONNX, TensorRT, PostgreSQL, MongoDB, Redis, Pinecone, GCP Vertex AI, Responsible AI, GDPR.

Professional Experience

AI Lead & Computer Vision Researcher — GraviLog (Remote, Part-time) | Sep 2025 – Present

- Defined AI roadmap and scalable ML architecture aligned with business goals.
- Led R&D initiatives in computer vision and multimodal AI systems.
- Conducted code reviews, mentored engineers, and optimized deployment pipelines.

Senior AI Developer — OMC Technologies (Remote) | May 2023 – Aug 2025

- Designed AI agents using Falcon, DeepSeek, and StarCoder to automate client workflows.
- Built multimodal healthcare AI systems, improving chatbot accuracy by 19%.
- Reduced API latency by ~30% using token streaming and optimized inference pipelines.
- Improved fraud detection accuracy by 17% for fintech applications.
- Deployed FastAPI and Docker microservices with sub-100ms latency and mentored 5 engineers.

AI Developer & Researcher — Nexgro (Krugersdorp, South Africa) | Apr 2020 – Apr 2023

- Developed computer vision models for grain quality analysis and crop monitoring.
- Built satellite/drone imagery pipelines for anomaly detection and yield prediction.
- Created NLP assistants for agricultural operations using LLM-based systems.
- Automated document intelligence workflows for contracts and silo receipts.
- Researched RL-based logistics optimization and predictive storage control systems.

Data Scientist — Nexgro (Krugersdorp, South Africa) | Mar 2018 – Mar 2020

- Built forecasting and route optimization models to reduce logistics costs.
- Developed predictive maintenance systems for agricultural equipment.
- Designed inventory optimization models to reduce spoilage and storage costs.

Systems Administrator — Egyptian Co. for Int. Trade | Oct 2010 – Mar 2018

- Managed IT infrastructure for 200+ users across 4 locations with 99.8% uptime.

Selected Research

Research focus includes Geometric AI, Physics-Informed Neural Networks, multimodal computer vision, uncertainty-aware segmentation, and efficient AI systems for deployment under limited compute and sparse data conditions.

Leadership & Mentorship

Mentored AI engineers, organized internal AI workshops, and led cross-functional AI initiatives.

Education

High Institute For Specialized Technological Studies – Egypt
Management and Information Systems Department | 2001 – 2005

Certifications

IBM Data Science, Deep Learning Specialization (DeepLearning.AI), TensorFlow Developer (Google), Advanced TensorFlow, Google Cloud MLOps, Research Methodology (University of London), CCNA 200-301, ENCOR 350-401.